

ABDULLAH SHAMAIL

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Ames, Iowa | Open to Relocation

EDUCATION

Iowa State University

Ph.D. Computer Engineering (GPA: 3.90/4.00)

Ames, IA
Expected May 2026

- **Thesis:** Algorithms and techniques for analyzing moving clusters in spatio-temporal data
- **Coursework:** Machine Learning, NLP, Big Data Tools, Algorithms, Software Systems

Iowa State University

M.S. Computer Engineering (GPA: 3.89/4.00)

Ames, IA
Dec 2024

LUMS

B.S. Computer Science (Dean's Honor List)

Lahore, Pakistan
May 2020

TECHNICAL SKILLS

ML / AI: TensorFlow, PyTorch, GNNs, Transformers, RNNs, GANs, LLM, LangChain, RAG

Big Data & Systems: PySpark, Apache Spark, Flink, Hadoop, SQL, Linux/Unix

Languages: Python, C++, Java, JavaScript, PHP, SQL

Tools: LaTeX, Git, MDAnalysis, NumPy, Pandas, Docker, Azure/AWS, scikit-learn

WORK EXPERIENCE

Iowa State University

Graduate Research Assistant

Ames, IA
Jan 2021 – Present

- Created **BARMC Algorithm (SIGSPATIAL 2024)**, delivering **1.7× faster** Hydrogen-Bond analysis for 800k-frame molecular simulations. *Awarded NSF Grant.*
- Developed **Trajectory Compression (ADBIS 2022)** framework achieving **80% storage reduction** and **3× faster** queries. *Received Best Paper Award.*
- Designed **SDMC framework (Pending 2026)** implementing spatial indexing to achieve **65× speedups** in diversity-aware clustering.
- Built pipelines for **motif mining** using geometric descriptors to identify aromatic ring-pair configurations for event prediction.
- Optimized Hadoop MapReduce algorithms for **Big Data Analysis**, achieving a **2× speedup** in text corpora processing.

Altair Engineering Inc.

Machine Learning Engineer Intern

Troy, MI
May 2025 – Aug 2025

- Engineered GNN and Transformer-based architectures for PhysicsAI models, achieving a **10× reduction in MAE** for spatio-temporal simulations.
- Deployed scalable pipelines using PyTorch and TensorFlow for GPU-accelerated inference in production systems.
- Optimized inference latency and memory profiling to support high-resolution physical predictions.

Taar Consulting

Software Engineer Intern

Lahore, Pakistan
May 2018 – June 2018

- Implemented a full-stack Customer Feedback Plugin for Retail Pro Prism using Python and JavaScript
- Captured feedback across **1,000+ daily transactions** to drive customer experience analysis.

TECHNICAL PROJECTS

AutoJobScout – AI Job Discovery Platform | *LangGraph, LangChain, RAG, LLMs, ETL, Streamlit*

2025

- Architected an **Agentic AI system** using **LangGraph** to orchestrate multi-agent workflows and **RAG pipelines**, achieving **92% matching accuracy** via semantic search and LLM-powered resume analysis.
- Engineered an **ETL pipeline** across 7 asynchronous microservices to process **1,000+ job listings daily**, optimizing inference via **batch processing** to reduce latency to **<15 seconds**.

Selected ML Projects | *PyTorch, Keras, TensorFlow*

2021 – 2022

- Built **LSTM** sarcasm classifiers (**84% precision**), **DCGAN** emoji synthesizers, and **CNNs** (CIFAR-10) for high-fidelity image classification.

LEADERSHIP & TEACHING

Iowa State University

Student Instructor / Teaching Assistant

Ames, IA
Jan 2021 – Present

- Mentored **400+ students** in Large Scale Data Analysis and Digital Logic
- Updated curriculum to focus on real-world applications in software safety and data management.

Pakistan Student Association (PSA)

President

Ames, IA
May 2023 – Apr 2024

- Raised **\$20,000** for flood relief via a social media campaign leveraging real-time analytics.